

MAHEE NOOR TAYBA

Austin, Texas

(254) · 715 · 0475 • mahee.noor.tayba@gmail.com • <https://tayba-mahee.github.io/>

SUMMARY

Data scientist and computer science professional with experience collecting, cleaning, validating, and analyzing large datasets to support research and data-driven decision-making. Proficient in Python, R, SQL, PostgreSQL, statistical modeling, machine learning, predictive analytics, ETL workflows, and data visualization. Experienced developing automated data pipelines, building predictive models, generating technical reports, and collaborating with cross-functional teams to deliver actionable insights.

EDUCATION

Master of Science in Computer Science, 2023

Baylor University, Waco, Texas.

Bachelor of Science in Computer Science and Engineering, 2017

Chittagong University of Engineering and Technology, Bangladesh.

TECHNICAL STRENGTHS

Languages	Python, R, SQL, Java, C++, JavaScript, HTML5
Data	Pandas, NumPy, Data Cleaning, Validation, Visualization, Statistical Analysis, Predictive Analytics
Databases	PostgreSQL, MySQL, SQL Query Optimization
Machine Learning	Scikit-learn, TensorFlow, PyTorch, Classification, Regression
Tools	Git, GitHub, Linux, Jupyter Notebook, Excel, REST APIs

EXPERIENCE

Adjunct Lecturer, Baylor University, Waco, Texas

Aug 2023 - Jan 2026

- Taught undergraduate courses in computer and microprocessor architecture and discrete mathematics.
- Developed assignments and assessments emphasizing analytical thinking, problem solving, and debugging.
- Communicated complex technical concepts to diverse student audiences.

Teaching Assistant, Baylor University, Waco, Texas

Jan 2022 - May 2023

- Assisted with programming and computer systems courses through labs and tutorials.
- Helped students troubleshoot software and system-level issues.
- Graded assignments and provided technical feedback to improve student performance.

Lecturer, City University & The Millennium University, Dhaka, Bangladesh

Feb 2018 - Dec 2020

- Served as a full-time lecturer teaching undergraduate computer science courses in the department of Computer Science with large class sizes (70–90 students).
- Taught courses including Data Structures, Advanced Java, System Analysis and Design, and Numerical Analysis, topics related to system design, architecture, and structured problem-solving.
- Designed course materials and assessments and supervised undergraduate capstone projects.
- Participated in curriculum development and departmental academic activities.

RESEARCH EXPERIENCE

Graduate Research Assistant, Baylor University, Waco, Texas

Jan 2021 - Dec 2022

- Designed and maintained data processing workflows for large-scale research datasets.
- Collected, cleaned, validated, transformed, and analyzed structured and unstructured data using Python.
- Developed automated ETL pipelines for data collection, transformation, validation, and analytical reporting.
- Performed statistical analysis and data quality validation to ensure accuracy and consistency across datasets.
- Designed, trained, evaluated, and validated predictive models using structured datasets.
- Collaborated with interdisciplinary stakeholders to define reporting requirements, evaluate outcomes, and communicate analytical findings.
- Built dashboards, analytical reports, and data visualizations to communicate findings to technical and non-technical stakeholders.

PROJECTS

Baylor Navigation System

- Designed and maintained PostgreSQL data models supporting campus navigation and resource management services.
- Developed automated data ingestion workflows for PostgreSQL databases and validated location datasets to improve data quality.
- Implemented backend services using Spring Boot and optimized SQL queries to improve performance and data accessibility.
- Built RESTful APIs for efficient data exchange between frontend and backend systems.
- Performed data validation, testing, and SQL query optimization to improve data accuracy, reliability, and application performance.

Data-Driven User Modeling

- Collected, cleaned, and analyzed behavioral datasets to identify how cultural differences influence information-seeking behavior.
- Designed and executed user studies, collecting, validating, and analyzing structured datasets from diverse participant groups.
- Applied statistical analysis and quantitative methods to identify trends in navigation behavior and information recall.
- Applied natural language processing and sentiment analysis techniques to extract insights from textual datasets.
- Translated behavioral insights into actionable design recommendations for improving user experience.
- Applied human-computer interaction principles, including Fitts' Law, to evaluate and optimize interface design.

Satellite Image Processing System (SRGAN)

<https://ieeexplore.ieee.org/abstract/document/9799082>

- Developed an image super-resolution system using GAN-based architectures for satellite imagery enhancement.
- Designed data processing and validation pipelines to ensure model accuracy and consistency.
- Optimized system performance when handling large-scale image datasets.
- Improved output quality through iterative testing and model refinement.

Traffic Data Analysis and Prediction System

<https://arxiv.org/abs/2509.03819>

- Processed and analyzed large-scale transportation datasets containing traffic and accident records.
- Developed ETL-style preprocessing pipelines for data cleaning, transformation, and feature engineering.
- Implemented data validation procedures to ensure dataset quality and consistency.
- Queried and transformed large datasets using SQL-based workflows to support predictive analysis.

- Applied predictive modeling, classification techniques, and statistical analysis to identify factors affecting traffic severity.
- Produced analytical reports and visualizations to support data-driven decision making and communicate model performance.

Hybrid Quantum-Classical Detection System

<https://rivas.ai/pdfs/tayba2022using.pdf>

- Designed a hybrid machine learning system integrating quantum circuits with classical deep learning models.
- Developed predictive classification models and evaluated performance across multiple datasets.
- Implemented workflows for model testing, validation, and performance evaluation.
- Conducted cross-platform integration across different computational environments.
- Analyzed performance trade-offs between classical and hybrid approaches.

CORE COMPETENCIES

- | | |
|---------------------------|-----------------------|
| · Predictive Analytics | · Machine Learning |
| · Statistical Modeling | · Data Cleaning |
| · SQL Development | · ETL Pipelines |
| · PostgreSQL | · Data Validation |
| · Data Visualization | · Workflow Automation |
| · Technical Documentation | · Research Analytics |